



LESSON 12.3

SOLVING PROBLEMS INVOLVING SURFACE AREA OF RIGHT CIRCULAR CYLINDERS

LESSON INTRODUCTION

MA.7.GR.2.1 Given a mathematical or real-world context, find the surface area of a right circular cylinder using the figure's net.

MA.7.GR.2.2 Solve real-world problems involving surface area of right circular cylinders.

LEARNING TARGETS

- I can draw the net of a right circular cylinder.
- I can solve problems involving the surface area of a right circular cylinder.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.

ESSENTIAL QUESTIONS

- What is the relationship between the circumference of the circle and the length of the lateral area of the cylinder?
- How do you find the surface area of a right circular cylinder?

LESSON NARRATIVE

Lesson 3 continues supporting student understanding of benchmark MA.7.GR.2.2. Students once again look at the connection between the circumference of the base to a dimension of lateral area. Students find the surface area of a right circular cylinder within a real-world context.

COMPONENT SUMMARY

12.3.1 WARM-UP (~5 MIN.)

Find dimensions using proportions

12.3.3 GUIDED PRACTICE (20 MIN.)

Practice finding surface area in real-world context

12.3.2 GUIDED INSTRUCTION (20 MIN.)

Find surface area of cylinders in real-world context

12.3.4 WRAP-UP (~5 MIN.)

Ticket Out the Door

RESOURCES

GLOSSARY

Key Terms:

- Surface area of a cylinder

Additional Terms:

- Lateral area

MATERIALS

- (Optional) Calculators
- Chart Paper
- (Optional) Markers

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle articulating the difference between lateral area and surface area.
- Students may inadvertently substitute radius for diameter and vice versa.

THINGS TO CONSIDER

- In questions that require several steps, students should not round until the final step as rounding mid-step could significantly change the answer. Consider using Desmos online scientific calculator.

LITERACY AND LANGUAGE ROUTINES

MLR: Clarify, Critique, Connect

Consider implementing this strategy by allowing students to trade their responses in 12.3.3 Guided Practice with another student. Students will then ask their partner clarifying questions as needed. This process will allow students to think more analytically and provide a more detailed mathematical explanation.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.6.1 Reasonableness

In 12.3.2 Guided Instruction students are required to perform multiple steps to find the surface area of right circular cylinders. Prompt students to review their answer for reasonableness as a strategy for checking their answers. Is it reasonable for the surface area of a push pop to be ~

56 cm? Using this critical thinking skill will help students mitigate incorrect responses.

DIFFERENTIATE INSTRUCTION

If students are struggling to find the surface area of right circular cylinders in 12.3.3 Guided Practice, consider providing a visual entry point by displaying an Anchor Chart model to reference when calculating the surface area. Also, provide students with an auditory entry point by having them discuss their strategies and answers with a partner.

12.3.1 WARM-UP: PHOTOGRAPHS

ENRICHMENT

What do you notice about the larger picture compared to the smaller pictures? What does it mean that the smaller picture is proportional to the larger picture?

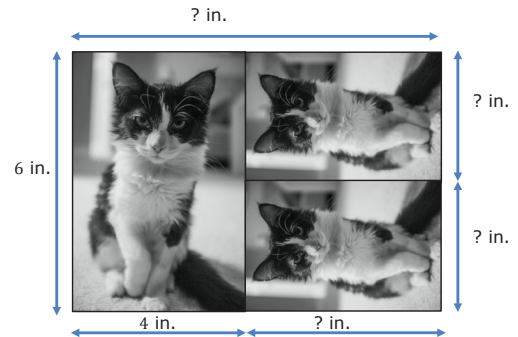
DIFFERENTIATE INSTRUCTION

Consider providing students with physical copies of photographs or images of the same dimensions to provide a kinesthetic entry point into this Warm-Up.



12.3.1 Warm-Up: Photographs

1. A photographer wants to print a photograph and two smaller copies on the same rectangular sheet of paper. The smaller picture is proportional to the larger picture. Dimensions are given in inches (in.). The diagram is not drawn to scale.



What are the dimensions of the small photographs?
3 in. by 4.5 in.

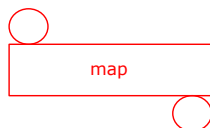


12.3.2 Guided Instruction: Finding the Surface Area of Right Circular Cylinders

- For her final art project, Courtney uses an old-world map to decorate a right-cylindrical pedestal. She wraps the map around the lateral area of the pedestal.



- Draw a net of the pedestal and label the part of the net where the map will be located.



- The length of the map is 16 feet (ft), and its height is 6.5 ft. The radius of the pedestal is 2.55 ft. Find the surface area of the pedestal. Use 3.14 for π and round your answer to the nearest hundredth.
 144.84 ft^2

- With your partner, write a formula that could be used to find the surface area of a right circular cylinder.

Answers will vary. $SA = 2\pi r^2 + \pi dh$

12.3.2 GUIDED INSTRUCTION: FINDING THE SURFACE AREA OF RIGHT CIRCULAR CYLINDERS

TEACHER MOVES

Think-Pair-Share

Consider utilizing an instruction routine to better position students as sense-makers during this activity.

- Students should work on each problem individually and record answers.
- Students will pair with their partner or group members to discuss solutions and provide constructive critiques.
- The teacher will select a few groups to share their solutions and methods.

- Ask two classmates to share and explain the formula they wrote. Record their formula in the table. *You are the only person who should write in the first column of the table.* Have the person verify that you correctly recorded their formula by writing their initials.

Formula	Initials
Answers will vary.	

- Review the formulas from your classmates with your partner. Determine if there is more than 1 formula that can be used to find the surface area of a right circular cylinder. If necessary, revise the formula you and your partner wrote.



Surface Area of a Cylinder

$SA = 2B + Ph$, where B is the area of the base, P is the perimeter of the base and h is the height
Or

$SA = 2\pi rh + 2\pi r^2$, where r is the radius of the circular base and h is the height

- The Push Pop candy, a creative spin on the lollipop, was very popular in the 1990s. It is cylindrical in shape, with a diameter of 2 centimeters (cm) and a height of 8 cm.



Find the surface area of the Push Pop candy. Use $\frac{22}{7}$ for π and round to the nearest hundredth.

56.57 cm^2

DIFFERENTIATE INSTRUCTION

Consider challenging students to discover a way to find the radius when given the length of the lateral area. Have students test their method to see if their way gives them the same radius in the problem.

12.3.3 GUIDED PRACTICE: SOLVING PROBLEMS INVOLVING SURFACE AREA OF RIGHT CIRCULAR CYLINDERS

DIFFERENTIATE INSTRUCTION

If students are struggling to find the surface area of right circular cylinders, consider providing a visual entry point by displaying an Anchor Chart model to reference when calculating the surface area.

TEACHER MOVES

Clarify, Critique, Connect

Consider implementing this strategy by allowing students to trade their responses with another student. Students will then ask their partner clarifying questions as needed. This process will allow students to think more analytically and provide a more detailed mathematical explanation.

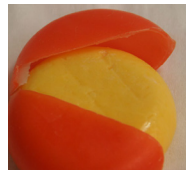
DIFFERENTIATE INSTRUCTION

Consider challenging students to discover a way to find the radius when given the length of the lateral area. Have students test their method to see if their way gives them the same radius in the problem.



12.3.3 Guided Practice: Solving Problems Involving Surface Area of Right Circular Cylinders

1. Cheese is often made into a cylindrical shape. Some cheeses are encased in paraffin wax after being formed into a cylindrical shape. The wax specially for coating cheeses aides in preventing unwanted mold growth and retains moisture. An example of what the wax coating looks like when removed from around the cheese is shown in the picture.



The circumference of a type of Gouda cheese shown is 9.5 in. and it has a height of 5 in.



How much wax coating will be needed to encase the cheese? Use $\frac{22}{7}$ for π and round your answer to the nearest hundredth.

61.86 in²

2. A checker piece is in the shape of a right circular cylinder and has a diameter of 1.75 in. Each checker piece that is sold in a vintage game set is individually wrapped snug in 6.9 in² of material. What is the height, in inches, of each individual checker piece? Use 3.14 for π and round your answer to the nearest hundredth.

0.38 in

3. Tim plans to give his wife a roll of toilet paper as a gag gift for their first wedding anniversary since paper is the traditional 1 year wedding anniversary gift. If the roll of toilet paper is 10 cm tall by 12 cm wide, how much wrapping paper, in cm², will he need to cover the gift? Use $\frac{22}{7}$ for π and round your answer to the nearest hundredth.

603.43 cm²



4. Amoré gave her homeroom teacher a cylindrical jar with a hot chocolate mix inside. The jar she used was 4.2 in. wide by 6 in. tall. The gift was such a hit that she wants to give her 5 other teachers the same hot chocolate mix and tightly wrap each jar in plastic to preserve the ingredients. How much plastic will she need in total to wrap the 5 jars? Leave your answer in terms of π and include units.

170.1 π in²

**12.3.4 Wrap-Up: Ticket Out the Door**

Consider the formula, $SA = 2\pi rh + 2\pi r^2$, to complete the following.

1. A hydration lotion is sold in a cylindrical container. The lotion ships in a box, but for quality control purposes the jar is wrapped in vacuum sealed plastic before being placed into a shipping box. The dimensions of the jar are 7.8 in. tall by 3.39 in. wide. How much plastic material is used to wrap around 60 individual jars of lotion? Use 3.14 for π and round your answer to the nearest whole number.

6064 in²

12.3.4 WRAP-UP: TICKET OUT THE DOOR**DIFFERENTIATE INSTRUCTION**

Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.